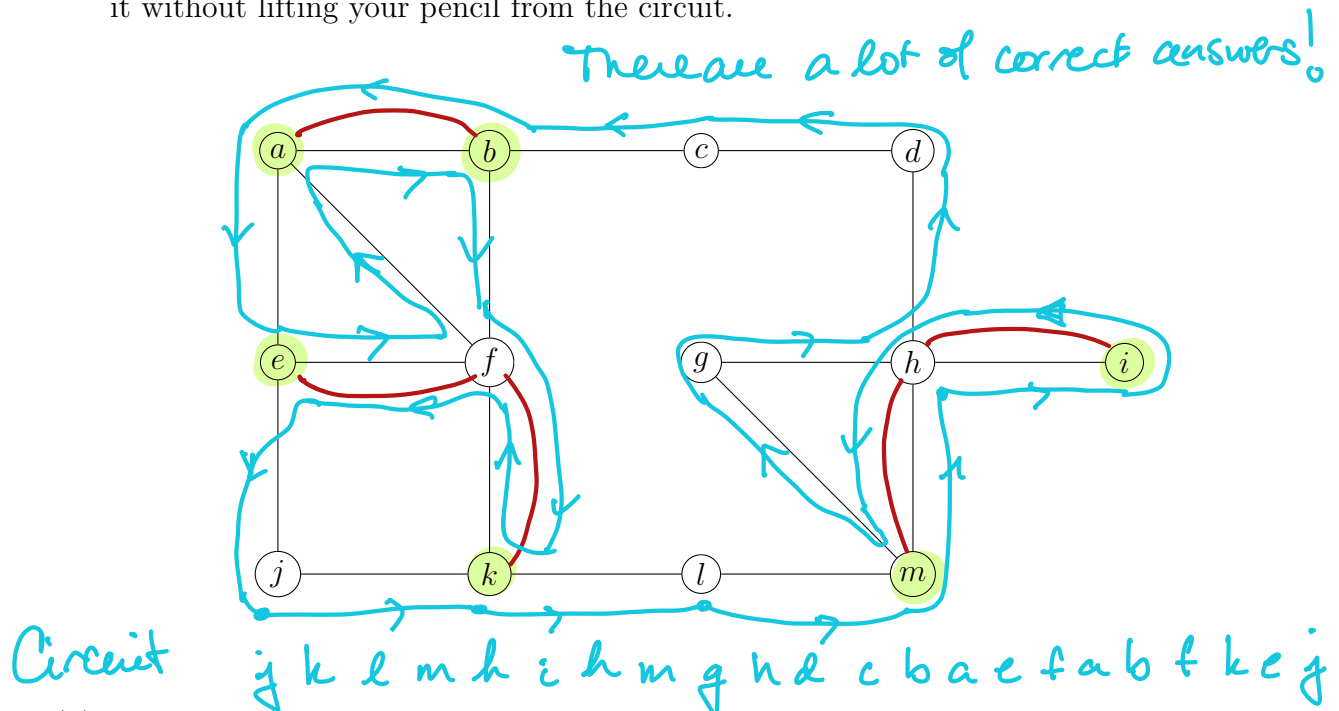


Worksheet: Eulerization

Group Names: Solutions

1. Consider the following graph.

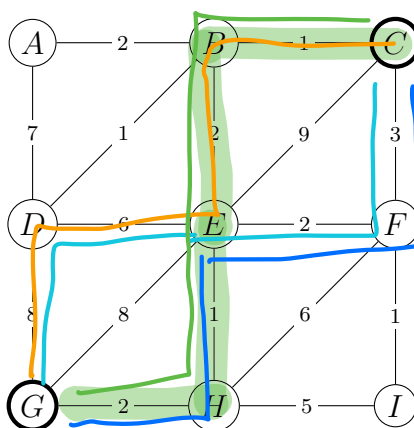
- (a) Which are the vertices of odd degree? a, b, e, k, i, m
- (b) **Eulerize this graph:** find the smallest number of edges you can duplicate so that you can construct an **Euler circuit**, and add them to the graph. 5 edges needed
- (c) Draw the circuit on the graph (offset slightly so you can see it) so that you can follow it without lifting your pencil from the circuit.



- (d) Give this graph a context: What might this graph represent? Why might you need an Euler circuit?

- Vertices are intersections of streets
- edges are roads
- Euler circuit is an efficient way to plow all the streets in the neighborhood, traversing already-plowed streets as few times as possible

2. Consider the following weighted graph.



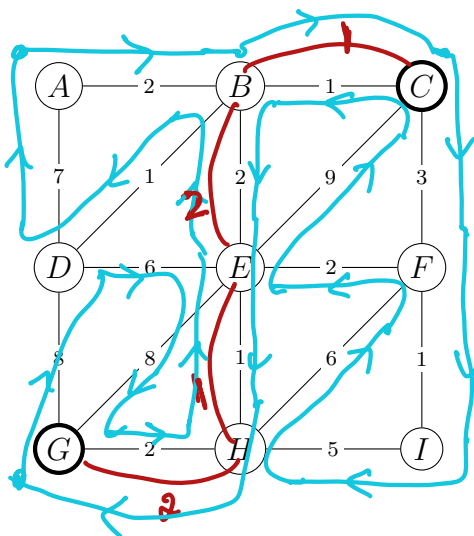
- (a) There are two vertices of odd degree in this graph, C and G . Use guess-and-check to find a path of shortest weighted distance between C and G . *Hint: the shortest-distance path has four edges.* (There is an algorithm that will do this, called Dijkstra's algorithm, which your textbook discusses — you can see why having a computer find shortest-distance paths might be useful!)

<u>Path</u>	<u>Wt</u>	<u>Path</u>	<u>Wt</u>
GEC	17	$GDEFC$	20
$GDEBC$	$8+6+2+1=18$	$GHEFC$	8
$GHEBC$	$2+1+2+1=5$		

What is your shortest-weighted-distance path? $GHEBC$

- (b) On the copy of the graph below, **duplicate your minimum distance path** (including the weights) to eulerize the graph. Then find an **Euler circuit** in the graph, which will be of minimum total weight.

The total weight of the circuit is the original weight sum plus the weight of the duplicated edges.



$ABC F I H F E C B E H G D E G E B D A$